

Problem

Let

$$R_1, R_2 \in \text{SO}(3).$$

Calculate

$$\boxed{\frac{\partial(R_1 R_2^{-1})}{\partial R_2}}$$

using both left and right perturbation models.

This note first derives the result for the *right perturbation model*. The left-perturbation derivation will be added separately.

1 Right perturbation

1.1 Method 1: Directly from the definition

Use the right-plus convention

$$R'_2 = R_2 \text{Exp}(\delta\theta), \quad \delta\theta \in \mathbb{R}^3.$$

Define

$$Y(R_2) = R_1 R_2^{-1}.$$

Since the output $Y \in \text{SO}(3)$ is also a Lie-group element, its difference is measured with the right-minus operator

$$Y' \ominus Y = \text{Log}(Y^{-1}Y').$$

To write the Jacobian limit rigorously, choose an arbitrary direction

$$u \in \mathbb{R}^3$$

and set

$$\delta\theta = hu, \quad h \rightarrow 0.$$

The right Jacobian is therefore defined by

$$\boxed{{}_{R_2}J_{R_2}^{R_1 R_2^{-1}} u = \lim_{h \rightarrow 0} \frac{Y(R_2 \text{Exp}(hu)) \ominus Y(R_2)}{h}}. \quad (1)$$

Substituting

$$Y(R_2) = R_1 R_2^{-1},$$

gives

$${}_{R_2}J_{R_2}^{R_1 R_2^{-1}} u = \lim_{h \rightarrow 0} \frac{\text{Log}[(R_1 R_2^{-1})^{-1} R_1 (R_2 \text{Exp}(hu))^{-1}]}{h}. \quad (2)$$

Using

$$(R_1 R_2^{-1})^{-1} = R_2 R_1^{-1}$$

and

$$(R_2 \text{Exp}(hu))^{-1} = \text{Exp}(-hu) R_2^{-1},$$

the argument of the logarithm becomes

$$(R_1 R_2^{-1})^{-1} R_1 (R_2 \text{Exp}(hu))^{-1} \quad (1)$$

$$= R_2 R_1^{-1} R_1 \text{Exp}(-hu) R_2^{-1} \quad (2)$$

$$= R_2 \text{Exp}(-hu) R_2^{-1}. \quad (3)$$

Hence

$${}^{R_2}J_{R_2}^{R_1 R_2^{-1}} u = \lim_{h \rightarrow 0} \frac{\text{Log}(R_2 \text{Exp}(-hu) R_2^{-1})}{h}. \quad (3)$$

Now use the conjugation identity

$$\boxed{R \text{Exp}(a) R^{-1} = \text{Exp}(\mathbf{Ad}_R a).} \quad (4)$$

Therefore

$$R_2 \text{Exp}(-hu) R_2^{-1} = \text{Exp}(-h \mathbf{Ad}_{R_2} u).$$

Substituting into (3),

$${}^{R_2}J_{R_2}^{R_1 R_2^{-1}} u = \lim_{h \rightarrow 0} \frac{\text{Log}[\text{Exp}(-h \mathbf{Ad}_{R_2} u)]}{h}.$$

Near the identity,

$$\text{Log}(\text{Exp}(v)) = v,$$

so

$${}^{R_2}J_{R_2}^{R_1 R_2^{-1}} u = -\mathbf{Ad}_{R_2} u.$$

Since u is arbitrary,

$$\boxed{{}^{R_2}J_{R_2}^{R_1 R_2^{-1}} = -\mathbf{Ad}_{R_2}.} \quad (5)$$

For $\text{SO}(3)$,

$$\mathbf{Ad}_R = R.$$

Therefore the final right-perturbation result is

$$\boxed{{}^{R_2}J_{R_2}^{R_1 R_2^{-1}} = -R_2.} \quad (6)$$

2 Left perturbation

Use the left-plus convention

$$R'_2 = \text{Exp}(\delta\theta) R_2.$$

2.1 Method 1: From equation (57)

From the right-perturbation result,

$$\boxed{{}^{R_2}J_{R_2}^{R_1 R_2^{-1}} = -R_2.}$$

Equation (57) converts a right Jacobian into the corresponding left Jacobian:

$$\boxed{{}^\varepsilon J_X^{f(X)} = \mathbf{Ad}_{f(X)} {}^X J_X^{f(X)} \mathbf{Ad}_X^{-1}.} \quad (57)$$

For

$$f(R_2) = R_1 R_2^{-1},$$

this gives

$${}^\varepsilon J_{R_2}^{R_1 R_2^{-1}} = \mathbf{Ad}_{R_1 R_2^{-1}}(-R_2) \mathbf{Ad}_{R_2}^{-1}.$$

For $\text{SO}(3)$,

$$\mathbf{Ad}_R = R, \quad \mathbf{Ad}_R^{-1} = R^{-1}.$$

Therefore

$${}^\varepsilon J_{R_2}^{R_1 R_2^{-1}} = (R_1 R_2^{-1})(-R_2) R_2^{-1} \tag{4}$$

$$= \boxed{-R_1 R_2^{-1}}. \tag{5}$$

2.2 Method 2: Directly from the definition

Choose an arbitrary direction

$$u \in \mathbb{R}^3$$

and let

$$\delta\theta = hu, \quad h \rightarrow 0.$$

Define

$$Y = R_1 R_2^{-1}.$$

Under the left perturbation,

$$R'_2 = \text{Exp}(hu) R_2.$$

Hence

$$Y' = R_1 (R'_2)^{-1} = R_1 (\text{Exp}(hu) R_2)^{-1}.$$

Using

$$(AB)^{-1} = B^{-1} A^{-1},$$

we obtain

$$(\text{Exp}(hu) R_2)^{-1} = R_2^{-1} \text{Exp}(-hu),$$

so

$$\boxed{Y' = R_1 R_2^{-1} \text{Exp}(-hu) = Y \text{Exp}(-hu)}.$$

Because the output perturbation is left/global, use the left-minus operator:

$$Y' \ominus_L Y = \text{Log}(Y' Y^{-1}).$$

Therefore

$${}^\varepsilon J_{R_2}^{R_1 R_2^{-1}} u = \lim_{h \rightarrow 0} \frac{\text{Log}(Y' Y^{-1})}{h} \tag{6}$$

$$= \lim_{h \rightarrow 0} \frac{\text{Log}(Y \text{Exp}(-hu) Y^{-1})}{h}. \tag{7}$$

Using the conjugation identity

$$Y \text{Exp}(v) Y^{-1} = \text{Exp}(\mathbf{Ad}_Y v),$$

we have

$$Y \operatorname{Exp}(-hu)Y^{-1} = \operatorname{Exp}(-h\mathbf{Ad}_Y u).$$

Thus

$${}^\varepsilon J_{R_2}^{R_1 R_2^{-1}} u = \lim_{h \rightarrow 0} \frac{\operatorname{Log}(\operatorname{Exp}(-h\mathbf{Ad}_Y u))}{h} \quad (8)$$

$$= -\mathbf{Ad}_Y u. \quad (9)$$

Since u is arbitrary,

$${}^\varepsilon J_{R_2}^{R_1 R_2^{-1}} = -\mathbf{Ad}_Y.$$

Finally, because

$$Y = R_1 R_2^{-1}$$

and $\mathbf{Ad}_Y = Y$ for $\operatorname{SO}(3)$,

$$\boxed{{}^\varepsilon J_{R_2}^{R_1 R_2^{-1}} = -R_1 R_2^{-1}.}$$

Final comparison

For the right perturbation

$$R'_2 = R_2 \operatorname{Exp}(\delta\theta_R),$$

$$\boxed{{}^{R_2} J_{R_2}^{R_1 R_2^{-1}} = -R_2.}$$

For the left perturbation

$$R'_2 = \operatorname{Exp}(\delta\theta_L) R_2,$$

$$\boxed{{}^\varepsilon J_{R_2}^{R_1 R_2^{-1}} = -R_1 R_2^{-1}.}$$